

Experimentation & Expansion Plan

Member	Roll No.	A2 Domain	A3 Role
Valeena Afzal	25I-8023	Image · Audio · Video	Improvement I + II Lead
Maryam Zafar	25I-8033	Poisson · SDF	Improvement III Lead + Bonus
Laiba Noor	25I-8035	Helmholtz · Wave Eq.	Improvement IV Lead + Datasets

0. Overview & Research Framing

Assignment 3 extends the SIREN reproduction (A2) into **original research** suitable for a conference or workshop paper (target: NeurIPS / ICLR workshop, or an AI/ML conference in Pakistan/Asia). The plan below proposes **four improvements**, an **additional dataset track**, and a **bonus experiment** — all designed to run within Google Colab free tier (NVIDIA T4, 15 GB VRAM, ~12 hr session).

Central Research Question:

Can SIREN's signal-fitting quality and PDE-solving ability be improved by (a) better activation design, (b) a hybrid frequency-aware loss, (c) cross-domain transfer to medical imaging, and (d) adaptive omega scheduling — and do these gains hold across multiple signal modalities on public benchmarks?

Rubric Item	How this Plan Covers It
Research Proposal (improvement types)	4 improvements spanning architecture, loss, data, efficiency
Original model comparison	All experiments include SIREN A2-baseline numbers
Reproduced baseline comparison	SIREN reproduced (A2) is Column 2 in every table
Proposed method(s)	4 distinct proposed variants, each with ablation
Additional Dataset (Must)	Kodak24 + BSD68 for image; LibriSpeech for audio; IXI Brain MRI
Bonus experiment	Medical image super-resolution on IXI MRI dataset

Improvement I — Adaptive omega Scheduling (Valeena)

Motivation

SIREN uses a fixed `omega_0=30` throughout training. Low `omega_0` limits high-frequency capacity; too high causes unstable gradients. Recent work (FINER, CVPR 2024) shows the frequency support of the network is bottlenecked by initialization. We propose **Cosine-Annealed omega Scheduling (CAOS-SIREN)**: start with `omega_0=10` (stable, low-freq), then anneal up to `omega_0=60` over training, letting the network first fit the global structure then refine high-frequency detail.

Implementation (Colab-safe)

- Modify the **SineLayer** class to accept `omega_0` as a mutable attribute.
- Add a scheduler: $\omega(t) = 10 + 50 * (1 - \cos(\pi * t / T)) / 2$ (cosine warm-up).
- Update `omega_0` every 500 steps. No extra GPU memory needed.
- Train on same 512x512 cameraman image + Kodak24 (24 images, <1 MB total).
- Log PSNR every 500 steps to TensorBoard CSV (no `summary_fn` needed).

Comparison Table (planned)

Model	Dataset	PSNR (dB)	Steps	Notes
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SIREN (paper)	Cameraman	~50.0	15K	Original
SIREN (A2 baseline)	Cameraman	40.37	10K	Our repro
CAOS-SIREN (ours)	Cameraman	TBD	10K	Expected +2-4 dB
SIREN (A2)	Kodak24	---	10K	New dataset
CAOS-SIREN (ours)	Kodak24	TBD	10K	Expected gain

Resources

- FINER paper (CVPR 2024): https://openaccess.thecvf.com/content/CVPR2024/papers/Liu_FINER_...
- Kodak24 dataset (24 PNG images, free): <https://www.kaggle.com/datasets/sherylmehta/kodak-dataset>
- SIREN Colab starter: https://colab.research.google.com/github/vsitzmann/siren/blob/master/explore_siren.ipynb
- BSD500 (already used in A2): <https://www2.eecs.berkeley.edu/Research/Projects/CS/vision/bsds/>

Improvement II — Frequency-Aware Hybrid Loss (Valeena)

Motivation

Standard SIREN minimizes L2 pixel loss, which treats all frequencies equally. The Fourier spectrum of natural images is dominated by low-frequency content, so the network fits those first and may under-represent high-frequency edges and texture. We propose adding a **Fourier Frequency Loss (FFL)** term: $L_{\text{total}} = L_2 + \lambda * ||\text{FFT}(\text{pred}) - \text{FFT}(\text{gt})||_1$, where λ is tuned via grid search over {0.01, 0.05, 0.1}. This adds zero trainable parameters.

1. Compute 2D FFT of predicted and ground-truth image patches inside the loss function.
2. Add L1 penalty on the FFT magnitude difference (focus on high-frequency bins).
3. λ warm-up: start at 0, linearly ramp to target λ over first 2K steps.
4. Evaluate on cameraman + BSD68 (68 greyscale test images, standard denoising benchmark).
5. Report PSNR, SSIM, and qualitative edge sharpness comparison.

Model	Dataset	PSNR	SSIM	Steps
SIREN (A2 baseline)	Cameraman	40.37	---	10K
SIREN + FFL ($\lambda=0.05$)	Cameraman	TBD	TBD	10K
SIREN (A2 baseline)	BSD68 avg	---	---	10K
SIREN + FFL ($\lambda=0.05$)	BSD68 avg	TBD	TBD	10K

Resources

- BSD68 test set: <https://github.com/claasmichele/CBSD68-dataset>
- Focal Frequency Loss (FFL) reference: <https://arxiv.org/abs/2012.12821>
- skimage.metrics for SSIM: already in your A2 environment
- torch.fft.fft2 — native PyTorch 2.x, no extra install needed

Improvement III — H-SIREN Activation + SDF on Complex Shape (Maryam)

Motivation

H-SIREN (Gao & Jaiman, arXiv Oct 2024) replaces the first-layer activation from $\sin(x)$ to $\sin(\sinh(2x))$. This one-line change widens the supported frequency set at initialization, making it easier to fit fine detail without overfitting. It is faster than FINER and WIRE, and a direct drop-in. We will evaluate H-SIREN vs. SIREN on Poisson reconstruction AND SDF learning on the Stanford Bunny (a richer shape than the synthetic sphere used in A2).

Implementation

1. **H-SIREN patch:** in `siren.py`, change first SineLayer activation to `torch.sin(torch.sinh(2*x))`. All other layers remain `sin(omega_0 * x)`. No other changes needed.

2. **Poisson**: re-run gradient supervision (your A2 code) with H-SIREN backbone. Compare PSNR: SIREN (A2: 33.56 dB) vs H-SIREN.
3. **SDF - Stanford Bunny**: download bunny.xyz (~3 MB, free). Run train_single_sdf.py with batch_size=1400 (safe for T4). Compare Eikonal satisfaction and mesh quality vs A2 sphere.
4. **Additional dataset**: run Poisson reconstruction on a SECOND image from BSD500 (e.g., the standard "peppers" image) to show cross-image generalization.

Model	Task	PSNR / Eikonal	Steps	Notes
SIREN (A2)	Poisson (cameraman)	33.56 dB	10K	A2 result
H-SIREN	Poisson (cameraman)	TBD	10K	Expected >34 dB
SIREN (A2)	Poisson (peppers)	---	10K	New image
H-SIREN	Poisson (peppers)	TBD	10K	New image
SIREN (A2)	SDF (sphere)	$ \text{grad} \sim 1$	5K	A2 result
H-SIREN	SDF (bunny)	TBD	5K	Richer geometry

Resources

- H-SIREN paper: <https://arxiv.org/abs/2410.04716>
- H-SIREN code: <https://github.com/RuiGao0213/H-SIREN>
- Stanford Bunny .xyz: <https://graphics.stanford.edu/data/3Dscanrep/> (or trimesh.creation + export)
- BSD500 "peppers" image: included in your A2 BSD500 dataset download

Improvement IV — k-Ablation on Helmholtz + Wave IC Variation (Laiba)

Motivation

The Helmholtz wavenumber k directly controls signal oscillation frequency. The original paper uses a single k value; no systematic study of how SIREN convergence and wavefield quality vary with k has been published. This is a genuinely open research question with physical relevance. Similarly, the wave equation's convergence depends on the initial displacement profile — a factor never studied in the original paper. Both experiments are entirely novel and publishable.

Helmholtz k-ablation

1. Identify the k parameter in train_helmholtz.py (line ~40, default $k=1$ or $k=3$).
2. Train 4 separate runs: k in $\{1, 5, 20, 50\}$. Each run: 5K steps (T4 safe ~63 min each).
3. Record: final loss, loss reduction %, wavefield sharpness (visual), convergence speed.
4. Plot: loss curves for all 4 k values on a single figure.
5. Hypothesis: SIREN handles large k better than ReLU-PE due to periodic activations.

Wave Equation Initial Condition (IC) Study

1. IC-1 (A2 baseline): existing default in train_wave_equation.py.
2. IC-2: Gaussian pulse — $u(x,0) = \exp(-|x|^2 / 0.05)$. Add to dataset class.
3. IC-3: Double sinusoid — $u(x,0) = \sin(2\pi x) + 0.5\sin(4\pi x)$.
4. For each IC: 5K steps, batch=3000, 256 hidden units (your A2 OOM fix).
5. Compare: loss reduction %, $u(x,t)$ spacetime solution quality.

Experiment	Config	Final Loss	Steps	Notes
Helmholtz $k=1$ (baseline)	A2 repro	175,564	5K	A2 result
Helmholtz $k=5$	New	TBD	5K	New
Helmholtz $k=20$	New	TBD	5K	New
Helmholtz $k=50$	New	TBD	5K	New

Wave IC-1 (A2 default)	A2 repro	16,236	5K	A2 result
Wave IC-2 (Gaussian)	New	TBD	5K	New
Wave IC-3 (Double sin)	New	TBD	5K	New

Resources

- train_helmholtz.py in SIREN official repo: https://github.com/vsitzmann/siren/blob/master/experiment_scripts/train_helmholtz.py
- train_wave_equation.py: https://github.com/vsitzmann/siren/blob/master/experiment_scripts/train_wave_equation.py
- PyTorch autograd for custom ICs: built-in (no extra install)
- Helmholtz equation reference (physics): Colton & Kress, "Inverse Acoustic and Electromagnetic Scattering", Springer

5. Additional Dataset Track (Required) — All Members

Three public, Colab-downloadable datasets are added across the three experiment domains. Each is small enough to fit in free Colab RAM and storage.

Dataset	Domain	Size	Member	Download / Notes
Kodak24	Image	24 PNGs ~1 MB	Valeena	kaggle datasets download -d sherylmehta/kodak-dataset
BSD68	Image	68 PNGs ~2 MB	Valeena	github.com/clausmichele/CBSD68-dataset
LibriSpeech test-clean sample	Audio	10 WAV ~5 MB	Valeena	openslr.org/12 — download test-clean, pick 10 clips
BSD500 "peppers"	Image/Poisson	<1 MB	Maryam	Subset of A2 BSD500 dataset
Stanford Bunny	SDF	~3 MB .xyz	Maryam	graphics.stanford.edu/data/3Dscanrep or trimesh export
IXI Brain MRI	Medical (Bonus)	~50 MB (axial slices)	Laiba	brain-development.org — IXI dataset, T1 modality, 2D slices

6. Bonus Experiment — SIREN for Medical Image Super-Resolution (Laiba)

Why this is publishable

Cross-domain evaluation of SIREN on medical imaging is an active 2024-2025 research area (MICCAI 2024 tutorial on INRs for medical imaging). Showing that SIREN — even in its basic form — can super-resolve brain MRI slices at 2x and 4x magnification, and that the omega schedule (Improvement I) further boosts performance, is a concrete novel contribution that could target a workshop paper or a local conference.

Method

1. **Dataset:** IXI Brain MRI, T1 modality. Download ~20 volumes. Extract 2D axial slices (256x256). Downsample by 2x and 4x to create LR inputs.
2. **Task:** train SIREN to map (x,y) -> pixel intensity. Train on LR slice, evaluate PSNR/SSIM of the SR output vs HR ground truth.
3. **Baselines:** bicubic interpolation (trivial), ReLU MLP (from A2), SIREN (A2 baseline).
4. **Proposed:** SIREN + FFL loss (Improvement II) + CAOS scheduling (Improvement I).
5. **Colab constraint:** use 20 slices x 2 magnifications = 40 training runs x ~3 min each = ~2 hours total. Very feasible.
6. **Metric:** PSNR, SSIM, visual comparison of fine grey-matter/white-matter boundaries.

Method	Scale	PSNR (dB)	SSIM	Notes
Bicubic	2x	---	---	Trivial baseline
ReLU MLP	2x	---	---	A2 baseline
SIREN (A2)	2x	---	---	Our A2 repro
SIREN + FFL + CAOS	2x	TBD	TBD	Our proposed
Bicubic	4x	---	---	Trivial baseline
SIREN (A2)	4x	---	---	Our A2 repro
SIREN + FFL + CAOS	4x	TBD	TBD	Our proposed

Resources

- IXI dataset: <https://brain-development.org/ixi-dataset/> (free, CC license)

- MICCAI 2024 INR tutorial: <https://inr4miccai.github.io/>
- Medical INR super-resolution reference: <https://arxiv.org/abs/2310.00490>
- torchio for MRI loading (pip install torchio): <https://torchio.readthedocs.io/>
- skimage.transform.resize for downsampling: in your A2 env

7. Google Colab Free-Tier Execution Plan

Key constraint: Colab free tier gives ~12 hr continuous GPU, T4 15 GB VRAM, ~77 GB RAM, ~107 GB disk. Sessions disconnect after ~90 min idle. **Strategy:** save checkpoints every 500 steps to Google Drive; resume from checkpoint if session disconnects.

Task	Member	Est. Time (T4)	Sessions Needed	Colab Risk
Imp I: CAOS image (5 omega values)	Valeena	5 x 7 hr = 35 hr	3-4 sessions	Medium — split runs
Imp I: Kodak24 (24 images)	Valeena	24 x 20 min = 8 hr	1 session	Low
Imp II: FFL loss (3 lambdas)	Valeena	3 x 7 hr = 21 hr	2 sessions	Medium
Imp III: H-SIREN Poisson x2 images	Maryam	2 x 40 min = 1.3 hr	<1 session	Low
Imp III: SDF bunny	Maryam	~2 hr	1 session	Low
Imp IV: Helmholtz k-ablation (4 runs)	Laiba	4 x 63 min = 4 hr	1 session	Low
Imp IV: Wave IC variation (3 runs)	Laiba	3 x 67 min = 3 hr	1 session	Low
Bonus: MRI super-res (40 slices)	Laiba	~2 hr	1 session	Low

Session Management Tips

Mount Google Drive at session start: `from google.colab import drive; drive.mount("/content/drive")`
Save checkpoint: `torch.save(model.state_dict(), "/content/drive/MyDrive/siren_ckpt_step{i}.pt")`
Resume: `model.load_state_dict(torch.load(path))` then continue training loop from saved step.
Use Colab Pro trial (\$10/month) for Improvement I only if free sessions are insufficient — NOT mandatory.
Valeena's local RTX 3050 can run Kodak24 and BSD68 image experiments (7 hr each) to save Colab quota.

8. Detailed Step-by-Step Work Division

#	Step	Owner	Deadline (suggested)
V1	Add omega scheduler to SineLayer; unit-test with a 1D signal	Valeena	Day 1
V2	Run CAOS-SIREN on cameraman (5 omega configs, 10K steps each)	Valeena	Day 2-3
V3	Download Kodak24; run SIREN + CAOS-SIREN on all 24 images	Valeena	Day 4
V4	Implement FFL loss (<code>torch.fft.fft2</code> + L1 on magnitudes)	Valeena	Day 3
V5	Grid-search lambda in {0.01, 0.05, 0.1} on cameraman	Valeena	Day 4-5
V6	Run best FFL config on BSD68 (68 images); collect PSNR/SSIM	Valeena	Day 5-6

V7	Plot: PSNR vs omega curve, FFL vs baseline PSNR histogram	Valeena	Day 7
M1	One-line H-SIREN patch in siren.py; verify gradient flow	Maryam	Day 1
M2	Run H-SIREN Poisson gradient-sup on cameraman (10K steps)	Maryam	Day 2
M3	Run SIREN + H-SIREN Poisson on BSD500 "peppers" image	Maryam	Day 2-3
M4	Download Stanford Bunny .xyz; run SDF (5K steps, batch=1400)	Maryam	Day 3-4
M5	Visualise SDF: marching cubes mesh + XY/XZ/YZ cross-sections	Maryam	Day 4
M6	Compare H-SIREN vs SIREN: Eikonal grad -1, mesh vertex count	Maryam	Day 5
M7	Prepare Improvement III figures and tables for report	Maryam	Day 7
L1	Find k parameter in train_helmholtz.py; parameterise via argparse	Laiba	Day 1
L2	Run Helmholtz for k in {1,5,20,50} — 4 x 5K step runs on T4	Laiba	Day 2-3
L3	Plot loss curves for all k values; note instability at k=50	Laiba	Day 3
L4	Implement IC-2 (Gaussian) and IC-3 (double-sin) in dataset class	Laiba	Day 3
L5	Run wave equation for IC-1, IC-2, IC-3 (3 x 5K steps)	Laiba	Day 4
L6	Download IXI MRI (T1); extract 20 axial slices, create 2x/4x LR	Laiba	Day 4
L7	Run bicubic / ReLU / SIREN / SIREN+FFL+CAOS on MRI (2x + 4x)	Laiba	Day 5-6
L8	Collect PSNR/SSIM for bonus; plot visual comparison (1 sample)	Laiba	Day 6
L9	Prepare Improvements IV + Bonus figures and tables	Laiba	Day 7

9. Master Resource List

SIREN official repo	github.com/vsitzmann/siren	All
SIREN official Colab	colab.research.google.com/github/vsitzmann/siren/...	All
FINER (CVPR 2024)	arxiv.org/abs/2312.02189	Imp I background
H-SIREN (Oct 2024)	arxiv.org/abs/2410.04716 + github.com/RuiGao0213/H-SIREN	Imp III
Focal Freq Loss	arxiv.org/abs/2012.12821	Imp II
Kodak24	kaggle.com/datasets/sherylmehta/kodak-dataset	Imp I dataset
BSD68	github.com/clausmichele/CBSD68-dataset	Imp II dataset

LibriSpeech	openslr.org/12 (test-clean)	Audio dataset
Stanford Bunny	graphics.stanford.edu/data/3Dscanrep/	Imp III SDF
IXI Brain MRI	brain-development.org/ixi-dataset/	Bonus
torchio (MRI loader)	<code>pip install torchio</code> torchio.readthedocs.io	Bonus
skimage.metrics (PSNR/SSIM)	already in A2 env after your fix	All metrics
torch.fft.fft2	built-in PyTorch 2.x, no install	Imp II FFL
MICCAI INR tutorial 2024	inr4miccai.github.io	Bonus context
TensorBoard CSV export	<code>tensorboard --logdir=./logs --export_dir=./csv</code>	All logging

10. A3 Report Structure (for paper submission)

1. Introduction	Motivate 4 improvements from A2 gaps. State research question.
2. Background	SIREN, spectral bias, FINER/H-SIREN, FFL. ~0.5 page.
3. Proposed Methods	Subsection per improvement. Equations, pseudocode.
4. Experimental Setup	Table: all datasets, hardware, hyperparameters.
5. Results	One subsection per improvement. Tables + figures. Compare to A2 baseline.
6. Bonus: Medical SR	IXI results. PSNR/SSIM table. Visual comparison.
7. Discussion	Which improvements help most? Where does SIREN still fail?
8. Conclusion	Summary of contributions. Future work.
9. References	Sitzmann 2020, FINER, H-SIREN, FFL, WIRE, IXI, BSD, Kodak.

Publication Target: NeurIPS 2025 Workshop on Neural Fields / ICLR 2026 Workshop on Implicit Neural Representations / ICASSP 2026 (audio/signal track) / local venues: ICET Pakistan, INMIC. All four improvements together form a coherent paper titled: *"Improving SIREN: Adaptive Frequency Scheduling, Spectral Loss, and Cross-Domain Evaluation of Implicit Neural Representations"*.

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